GR2

THE CLIMB SPACE

POVS, HMWS, BRAINSTORMING,
& EXPERIENCE PROTOTYPING

GR2

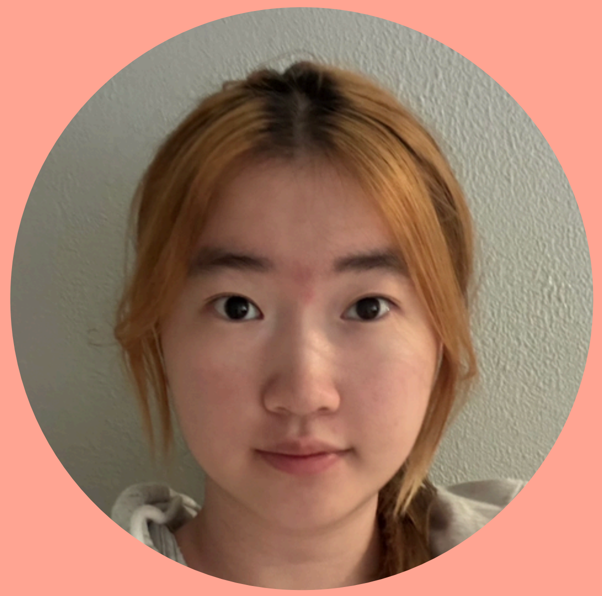
TEAM MEMBER

MEET THE TEAM

Hyun Cho

MAT

Korea, Seoul



Shuang Li

CS

China

Xandra Wong

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Malaysia, KL

ORIGINAL DOMAIN SELECTION

HOW DID WE ARRIVE AT BOULDERING?

WHAT INTERESTING CHALLENGES DO PEOPLE FACE?

CAN AI BE USED TO IMPROVE IT?

COULD ACCESSIBILITY BE IMPROVED HERE?

WHAT ACTIVITY INVOLVES LEARNING, DECISION-MAKING, AND SOCIAL INTERACTION?

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REFINED DOMAIN

ACCESS TO GUIDANCE AND ENCOURAGEMENT TO ALL TYPES OF CLIMBERS WHO NEED IT

CLIMBING TERMS USED IN THIS STUDY

STYLE (MOVEMENT STYLE)
THE TYPE OF MOVEMENT A ROUTE REQUIRES, SUCH AS DYNAMIC JUMPS, BALANCE MOVES, OR STRENGTH-BASED PULLING. DIFFERENT CLIMBERS PREFER AND PERFORM BETTER IN DIFFERENT STYLES.

ROUTE/PROBLEM
A SPECIFIC CLIMBING PATH SET ON THE WALL USING DESIGNATED COLORED HOLDS.

BETA
A CLIMBER'S STRATEGY OR SEQUENCE OF MOVES FOR FINISHING A ROUTE.

- **Recruitment channels:** visiting climbing gym / friends of friends
- **Criteria:** To apply feedbacks and for diversity, this time we added an expert level climber who enjoys climbing alone. Non-UCSB students; climbs regularly; variety in experience level; mix of body types

FINDING NEW PARTICIPANTS

ADDITIONAL INTERVIEWEES

REILY

27-YEAR-OLD
CLIMBING INSTRUCTOR

CHARLIE

22-YEAR-OLD
COLLEGE STUDENT

*names are pseudonyms

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INTERVIEW RESULTS

REILY
LOCATION: UCSB REC CENTER
USER TYPE: EXTREME USER
INTERVIEWER: HYUN
RECORDER: XANDRA
SCRIBE: SHUANG

Riley is an experienced climber and a climbing instructor at the UCSB Recreation Center. While she enjoys the communal aspect of climbing, Riley also values climbing alone, using that time to focus, reflect, and better understand her movement. She is highly aware of her own physical abilities and limitations, and prefers to explore solutions independently before seeking advice. Rather than relying immediately on external input, she chooses to ask for guidance only when she feels it is truly necessary.

"Even when I watch someone else, I still have to figure out my own way."

"It's better when people ask if I want advice."

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EMPATHY MAP

REILY
LOCATION: UCSB REC CENTER
USER TYPE: EXTREME USER
INTERVIEWER: HYUN
RECORDER: XANDRA
SCRIBE: SHUANG

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EMPATHY MAP

- Autonomy
- Self-awareness
- Reflective orientation

RILEY

SAYS

"I do enjoy climbing alone a lot."

"I feel like climbing alone is more based on me trying to be more technical and intentional with how I'm climbing."

"Sometimes I just want to be kind of be with myself while I'm climbing."

"I think climbing alone can be really relaxing."

"A lot of times it turns into a group effort."

"Sometimes I just need space."

"Some things are just physically harder for me."

"Sometimes they're just taller or stronger."

"I can't always do it the same way."

"Even when I watch someone else, I still have to figure out my own way."

"I'm really bad at crimps, but I'm good at pinches."

"I don't think failing means I'm bad at climbing."

"I think I've learned what my strengths and weaknesses are."

"It's not always about strength." "Sometimes it's more about body position."

"I think about whether it's possible for people of different heights."

"I think a lot about how routes are set."

"It's better when people ask if I want advice."

"Sometimes I see issues in the set rather than my own climbing."

"I think about whether a route is fair."

"Some routes are just harder for certain people."

"Everyone has different advantages or strengths that I might not have."

"Sometimes people give advice that just doesn't work for my body."

"The timing and tone really matter."

THINKS

Everyone has different advantages or strengths that I might not have.

The same approach does not work for every climber.

Failure is not necessarily a lack of ability, but sometimes a result of the conditions.

I may need help, but I want to stay in control of when and how I receive it.

Being alone is not avoidance, but a way of thinking.

It's not necessarily that I can't do it.

I know my strengths and weaknesses

I am both a participant and a setter."

Climbing does not have a single correct solution

Advice is useful only when it fits my body and when I'm ready to receive it

Climbing is closer to understanding than achievement.

I can try several times if I can't solve it this time

"Being alone while climbing helps me focus and understand my movement more clearly."

Not being able to do a route does not mean I should stop trying.

My body sets the conditions for how I climb.

Understanding comes from trying multiple times, not from getting it right once.

What matters is not completing the route, but understanding why it works or doesn't.

DOES

Tries routes through repeated attempts rather than planning in advance.

Adjusts foot placement and body position when something doesn't work.

Observes how others climb but does not copy their movement exactly.

Experiments independently before asking for help.

Takes breaks and returns to problems later with a fresh perspective.

Chooses to climb alone at times in order to focus and move intentionally.

Engages with others naturally when sharing a route or space.

Evaluates routes not only as a climber but also from a setter's perspective.

Stops attempting a route when it feels physically mismatched rather than forcing it.

Pays attention to body mechanics, reach, and positioning during movement.

Chooses not to engage in conversation while actively climbing.

Reflects on climbs afterward rather than tracking performance numerically.

Avoids relying heavily on apps or recording while climbing.

Engages socially when invited but does not initiate interaction while focused.

Uses observation rather than verbal explanation to learn movement.

Notices when a route favors certain body types or heights.

Go to gym outside of her workplace to have alone time

Set up new route every month

Climbs both alone and with others depending on mood and context.

FEELS

Feels mentally relaxed when not being observed or interrupted.

Feels more focused and calm when climbing alone.

Feels discouraged when others assume something is "easy."

Feels frustrated when a route feels physically mismatched to their body.

Feels annoyed when advice is given without being asked.

Feels uncertain whether failure is from personal problem or route problem

Feels mentally engaged when analyzing movement and body positioning.

Feels satisfied when understanding why a move does or does not work.

Feels more confident when figuring things out independently.

Feels a quiet sense of progress through small improvements.

Feels more comfortable when interactions happen naturally rather than forced.

Feels pressure when others watch or comment during attempts.

Feels less motivated when success is defined only by grades or completion.

Feels grounded when climbing becomes repetitive and focused.

Feels temporarily stuck but not defeated when failing on a route.

Feels temporarily stuck but not defeated when failing on a route.

Observed ⟷ Inferred

"It's better when people ask if I want advice."

INSIGHT

- Riley is an autonomy-driven climber who values independence in how she approaches problems.
- As an advanced climber, she does not only engage as a participant but also thinks from the perspective of a route setter.
- She believes that different climbers require different solutions, rather than a single correct way to climb.
- She has a strong awareness of her own physical abilities and limitations.
- Rather than receiving constant guidance, she prefers to seek advice only when it is necessary and relevant to her situation.
- While she enjoys climbing as a social activity, she also values climbing alone as a way to step away from her role and focus on herself.
- Riley values understanding the cause of failure more than achieving immediate success.

NEEDS

- She needs a way to seek advice without interrupting her personal focus.
- She needs a way to evaluate whether her routes work for climbers with different bodies.
- She needs a platform to receive feedback on her route design.
- She needs space to experiment with multiple solutions rather than follow fixed answers.
- She needs a way to reflect on why something worked or failed.
- She needs control over when and how social interaction happens.
- She needs moments where she can step out of her instructor role and simply explore.

"It's better when people ask if I want advice."

CHARLIE
LOCATION: ZOOM (LONDON)
USER TYPE: AVERAGE USER
INTERVIEWER: XANDRA
RECORDER: HYUN
SCRIBE: SHUANG

Charlie is an intermediate climber who has been climbing occasionally since 2022. She climbs once every 2 weeks, and was brought into climbing when her friends introduced her to it. She has been enjoying it as a fun, mentally engaging activity since.

"It's most frustrating when there's only one beta, and it doesn't suit you"

"The friends you go with have to be supportive and not judgemental."

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EMPATHY MAP

CHARLIE
LOCATION: ZOOM (LONDON)
USER TYPE: AVERAGE USER
INTERVIEWER: XANDRA
RECORDER: HYUN
SCRIBE: SHUANG

CHARLIE

SAYS

- throughout the process, you think a lot about how do I solve the puzzle, which is the fun aspect
- I look at it first and imagine myself using different angles to approach the holes
- I also look at other people to get inspiration
- might have to use cool moves that I can't think of
- I have to come down and think for a while
- If it's a very tough one, like beyond my comfort zone, then yes (watches beta), but otherwise I'll do it on my own
- some would consider watching beta as cheating
- I think because I'm in Europe, climbs are way longer
- By the third or fourth time, if I still fail, I ask for help or watch others
- or I might move on and try again next time
- Am I took weak for this? Which muscles am I not activating?
- When I go with friends, then everyone can do it but I can't then it's like...
- most frustrating when there's only one beta, and it doesn't suit you
- or if it's the last move and it takes a big leap of faith
- the friends you go with have to be supportive and not judgemental
- it's demotivating when you go alone, and there's no one to help you improve
- "aw man, all that effort" (when friends help her but still can't succeed)

THINKS

- I don't have much strength so I have to be wise about how I use it
- some moves are just beyond my thinking capacity
- I value self-reliance and learning by doing
- I have to work around climbs even if they don't suit my physique
- asking for help is not an embarassing to do
- I don't want to waste time and energy trying to do something on my own
- How do I improve? What am I doing wrong?
- Your judgement will kill my motivation
- I don't want to take the risk of getting injured, especially since I'm in a foreign coutnry
- why am i the only one who can't do it?
- Should I give up and try again next time?
- i'm too weak
- I need some help
- I can't do this on my own
- I don't want to let them down

DOES

- climbs once every 2 weeks
- only does bouldering, but switches between gyms
- occasionally internalizes failure, but also understands contextualizes it
- not afraid to seek help
- doesn't dwell on the same problem if it's not working
- tries again with full strength and a clear mind
- sad and demotivated expression when saying she's the only one out of the group who failed
- evaluates if certain moves are possible for her
- tries to solve it on her own before asking for help
- doesn't try the move if it's likely to lead to injury
- goes beyond her limit for the people who helped her

FEELS

- enjoys the idea of having new and unexpected routes by visiting different gyms
- feels very accomplished with being able to solve tough problems on her own
- confident in her abilities
- comfortable using betas as a resource
- sad and unsatisfied due to failure
- slightly worried about others' perception on her for watching betas
- prioritizes improvement over appearance
- prioritzes efficiency, both in energy and time
- isolated due to comparison
- afraid of falling and injuring herself
- values self autonomy
- strong frustration due to no progression and guidance
- ambitious, but not careless
- grateful for help, don't want it to go to waste

Observed ← → **Inferred**

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EMPATHY MAP

- Self-awareness
- Values supportive help

AYS

throughout the process, you think a lot about how do I solve the puzzle, which is the fun aspect

I look at it first and imagine myself using different angles to approach the holes

I also look at other people to get inspiration

might have to use cool moves that I can't think of

I have to come down and think for a while

If it's a very tough one, like beyond my comfort zone, then yes (watches beta), but otherwise I'll do it on my own

some would consider watching beta as cheating

I think because I'm in Europe, climbs are way longer

By the third or fourth time, if I still fail, I ask for help or watch others

or I might move on and try again next time

Am I took weak for this? Which muscles am I

When I go with friends, then everyone can do it but I can't n it's like...

most frustrating when there's only one beta, and it doesn't suit you

or if it's the ast move and it takes a big leap of faith

the friends you go with have to be supportive and not judgemental

it's demotivating when you go alone, and there's no one to help you improve

"aw man, all that effort" (when friends help her but still can't succeed)

THINKS

I don't have much strength so I have to be wise about how I use it

some moves are just beyond my thinking capacity

I value self-reliance and learning by doing

I have to work around climbs even if they don't suit my physique

asking for help is not an embarassing to do

I don't want to waste time and energy trying to do something on my own

How do I improve? What am I doing wrong?

Your judgement will kill my motivation

I don't want to take the risk of getting injured, especially since I'm in a foreign coutnry

why am i the only one who can't do it?

Should I give up and try again next time?

i'm too weak

I need some help

I can't do this on my own

I don't want to let them down

CHARLIE

DOES

climbs once every 2 weeks

only does bouldering, but switches between gyms

occasionally internalizes failure, but also understands contextualizes it

not afraid to seek help

doesn't dwell on the same problem if it's not working

tries again with full strength and a clear mind

sad and demotivated expression when saying she's the only one out of the group who failed

evaluates if certain moves are possible for her

tries to solve it on her own before asking for help

doesn't try the move if it's likely to lead to injury

goes beyond her limit for the people who helped her

FEELS

enjoys the idea of having new and unexpected routes by visiting different gyms

feels very accomplished with being able to solve tough problems on her own

confident in her abilities

comfortable using betas as a resource

sad and unsatisfied due to failure

slightly worried about others' perception on her for watching betas

prioritizes improvement over appearance

prioritzes efficiency, both in energy and time

isolated due to comparison

afraid of falling and injuring herself

values self autonomy

strong frustration due to no progression and guidance

ambitious, but not careless

grateful for help, don't want it to go to waste

Observed ⟷ Inferred

"It's most frustrating when there's only one beta, and it doesn't suit you"

INSIGHT

- Charlie values the satisfaction of solving tough problems and tries independently first, balancing effort with her awareness of strength and stamina limits.
- She prioritizes safety, avoiding risky moves that could cause serious injuries.
- Charlie prefers personalized, achievable solutions over general, difficult ones that are impossible to achieve.
- Receiving help motivates her to put in extra effort, reflecting her sense of reciprocity and social responsibility.

"It's most frustrating when there's only one beta, and it doesn't suit you"

NEEDS

- She needs ways to make progress without removing her sense of ownership in problem solving.
- She needs personalized help that considers her safety and physical capabilities.
- She needs encouragement to push her limits in a way that feels supportive rather than pressured or risky.

PREVIOUS KEY LEARNINGS

- Most climbers think the shared betas (example solutions) are designed for a standard body that doesn't match theirs
- Most climbers enjoy solving problems with others, not solitarily, though some feel pressured or discouraged by their advice
- When climbers get stuck, they often think it's personal failure rather than learning process
- Climbers want to visualize their progress

NEW KEY LEARNINGS

- Understanding one's own strength and weaknesses is essential for making progress and developing personal climbing strategies.
- Many climbers prefer to attempt a problem on their own first, and seek a beta they are physically able to follow after many failures. This helps preserve their sense of effort and accomplishment. Receiving guidance after struggling is more meaningful than being given the solution upfront.
- Some climbers value solitude, eventhough climbing is often seen as a communal activity

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POVS + HMWS

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SELECTED POV 1

- **WE MET...**

June, an 18-year-old UCSB freshman who is a casual-serious climber, enjoys bouldering as a social and problem-solving activity.

- **WE WERE SURPRISED TO NOTICE...**

She actively learns by watching and using shared betas, but many betas still don't work for her. And she is more likely to stop projecting when a climb feels outside her movement style.

- **WE WONDER IF THIS MEANS...**

She needs help recognizing that a beta mismatch is not personal failure, and needs to explore alternative ways that fit her body and style.

- **IT WOULD BE GAME CHANGING TO...**

Help climbers see and compare multiple body-appropriate movement options on the same problem.

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HMW

HOW MIGHT WE?

How might we help climbers explore alternative approaches when the common beta doesn't fit their body or style?

How might we reduce the assumption that failure is caused by body limitations rather than the beta itself?

How might we encourage climbers to adapt betas instead of copying them exactly?

How might we support climbers in trusting their own movement choices when others' betas don't work?

How might we help climbers feel included even when they can't follow the "expected" solution?

How might we help climbers recognize that multiple valid solutions can exist for the same problem?

How might we prevent beta-watching from turning into discouragement?

How might we reframe beta as inspiration rather than instruction?

How might we make non-standard or less visible betas easier to discover?

How might we help climbers experiment with body-specific strategies more confidently?

How might we help climbers learn from failed beta attempts instead of just repeating them?

How might we make beta differences visible without making climbers feel compared or judged?

SELECTED HMW1

How might we help climbers explore alternative approaches when the common beta doesn't fit their body or style?

SELECTED POV 2

- **WE MET...**

  Dan, a researcher from Belgium, who is a beginner-to-intermediate climber with one and a half years of experience. He enjoys climbing as a fun hobby instead of as an intense workout

- **WE WERE SURPRISED TO NOTICE...**

  that supportive advice, encouragement, and casual conversation during breaks noticeably increase his motivation and stamina more than climbing alone

- **WE WONDER IF THIS MEANS...**

  that climbing is as mentally demanding as it is physical, and social support plays a key role in sustaining effort

- **IT WOULD BE GAME CHANGING TO...**

  support climbers' mental energy during sessions, not just their physical performance

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HMW

How might we help climbers sustain motivation across an entire session, not just during successful climbs?

How might we give climbers specific advice when they climb, even when they climb alone?

How might we make mental recovery as intentional as physical rest in bouldering sessions?

How might we use breaks to improve motivation instead of brewing discouragement?

How might we enhance the effects of social support?

How might we reduce the mental demands of climbing?

HOW MIGHT WE?

How might we ensure climbers feel mentally supported during sessions, even if they don't have friends to go with?

How might we surface small wins that help climbers feel energized instead of drained?

How might we help climbers regain confidence after repeated failed attempts?

How might we ensure that climbers are always surrounded by supportive peers?

How might we recreate the social support and encouragement climbers get without requiring constant social presence?

How might we reduce the mental fatigue that builds up between attempts on difficult problems?

How might we make climbing feel less helpless and isolating when a climber trains alone?

SELECTED HMW 2

How might we ensure climbers feel mentally supported during sessions, even if they don’t have friends to go with?

SELECTED POV 3

- **WE MET...**

Riley, an experienced climber and climbing instructor at the UCSB Recreation Center. She has been climbing for several years and is deeply familiar with both climbing techniques and route setting.

- **WE WERE SURPRISED TO NOTICE...**

Even though Riley enjoys climbing with others, she intentionally goes to a more distant gym in order to have time to climb alone.
She approaches climbing not only as a participant, but also from the perspective of a route setter.
She does not see failure as a personal limitation, but as something influenced by body type, technique, or route design.

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SELECTED POV 3

- **WE WONDER IF THIS MEANS...**

Riley values uninterrupted focus and intentionally seeks distance from social interaction in order to think and reflect.

Advanced climbers may prioritize autonomy before seeking feedback.

Choosing when and how to receive advice is especially important at higher skill levels.

Not all climbers benefit from the same type of feedback.

At an advanced level, growth often comes from reflection and self-guided problem solving rather than direct instruction.

Route setters are already aware that not all problems work for every body type or skill level.

- **IT WOULD BE GAME CHANGING TO...**

Riley could receive advice only when she chooses, without disrupting her focus.

She could gather feedback on the routes she sets without constant in-person interaction.

She could understand how different types of climbers approach her routes.

She could move fluidly between the roles of instructor and learner within the same climbing space.

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HMW

How might we enable climbers who prefer climbing alone to receive guidance without disrupting their focus?

How might we allow climbers to choose when and how they receive advice?

How might intermediate and advanced climbers receive meaningful feedback that matches their skill level?

How might guidance be integrated into climbing without becoming a distraction?

How might route setters understand how climbers with different body types and skill levels experience their problems?

How might climbers learn through exploration rather than instruction?

HOW MIGHT WE?

How might we enable climbers who prefer climbing alone to receive guidance without disrupting their focus or autonomy?

How might climbers move beyond their habitual thinking and try new movement strategies?

How might introverted climbers seek advice without direct social interaction?

How might climbers receive feedback without interrupting their concentration or flow state?

How might climbers receive support even when an instructor is not present?

How might we collect and visualize different climbers' interpretations of the same route?

How might climbers go beyond surface-level tips and gain insights that support deeper learning?

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SELECTED HMW 3

HMW3: How might we enable advanced climbers who prefer climbing alone to find creative solution for questions without disrupting their autonomy?

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SOLUTIONS

CLIMBSPACE

HMW1: How might we help climbers explore alternative approaches when the common beta doesn't fit their body or style?

SOLUTION 1:

AN AI-ASSISTED SYSTEM THAT ANALYZES A CLIMBER'S FAILED ATTEMPT AND SUGGESTS MULTIPLE ALTERNATIVE BETAS BASED ON THEIR BODY TYPE, REACH, AND SKILL LEVEL.

INSTEAD OF PRESENTING A SINGLE "CORRECT" BETA, THE SYSTEM VISUALIES SEVERAL POSSIBLE MOVEMENT STRATEGIES THAT THE CLIMBER COULD REALISTICALLY ATTEMPT, HELPING THEM UNDERSTAND WHY A BETA DOES OR DOES NOT WORK FOR THEIR BODY.

SOLUTION-1

POV1

An AI-assisted system that analyzes a climber's failed attempt and suggests multiple alternative betas based on their body type, reach, and skill level.

PROTOTYPE

Critical Assumption:
When climbers understand why a beta does not work for their body, they feel less frustration and are more willing to try alternative movement strategies instead of blaming their ability.

Why critical:
If users do not perceive meaningful differences between suggested betas, the system fails to reduce frustration or improve learning.

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Prototype format

This prototype simulates an AI-powered climbing assistant.
Instead of using an actual AI system, a researcher takes on the role of the AI to generate responses.

Materials

- Video of participant's failed climb
- Self-reported body profile(height, reach, skill level)
- Annotated route image generated by "AI"

Target users— Someone who has different height from average male.

Procedures

Round 1 — User Input
Participants provide:
A photo of the climbing route they failed
A brief description of where they failed
Basic body information (height, reach, experience level)

Round 2 — "AI" Interpretation
A researcher acts as the AI system
Analyzes the mismatch between:

- the climber's body
- the original beta
- the route structure

Generates alternative beta paths manually

Round 3 — Visual Feedback
The new beta is drawn directly onto the route photo
Multiple possible paths may be shown
Each path reflects a different body strategy

Round 4 — User Reflection
Participants are asked:
Does this new beta feel more achievable?
Does it change how you interpret your failure?
Would this make you want to try again?

EXPERIENCE PROTOTYPE 1: PARTICIPANTS

Participant 1: Climber needing help

Who are they?
A female UCSB undergraduate student who is an intermediate climber.

How were they recruited?
At UCSB Recreation Centre's Climbing Gym

Why are they relevant?
She just came down from a climb after not being able to climb up and seemed like they would need help.

Participant 2: AI actor

Who are they?
A male UCSB climbing center staff who is an advanced climber.

How were they recruited?
At UCSB Recreation Centre's Climbing Gym

Why are they relevant?
He is a professional male climber who was taller and always helps others when people are in troubles.

EXPERIENCE PROTOTYPE 1: RESULT

What worked

- We simulated the AI assistant using a human “AI” (a staff climber) to generate alternative beta suggestions.
- The participant understood the idea of comparing body mismatch vs. beta mismatch.

What didn’t work

- A human cannot fully act like an AI system.
- The staff’s suggestions were still based on his own climbing style and reach.
- Expert beta is not automatically body-fit beta.

Was the assumption valid?

- Partially.
- The climber still could not complete the move.
- The suggested beta did not work for her body constraints.

New Learnings

- Our prototype suggest that body-aware beta must be generated from constraints, not just expertise.
- This supports why an AI-style multi-path generator is needed instead of single-expert advice.

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Participant 1

Participant 2

fail point

HMW2: How might we ensure climbers feel mentally supported during sessions, even if they don't have friends to go with?

SOLUTION 2:

CLIMBERS CAN "REQUEST BETA" SO NEARBY CLIMBERS GET A GENTLE NOTIFICATION, PROMPTING EXPERIENCED CLIMBERS TO APPROACH AND PROVIDE HELP, ELICITING CONVERSATION

SOME CLIMBERS MIGHT FEEL SHY TO ASK FOR HELP. HAVING AN APP THAT ALLOWS THEM TO ASK FOR HELP WITHOUT THEM HAVING TO DIRECTLY MAKE CONVERSATION WOULD EASE THEIR ANXIETY, AND DIRECTLY HELP THEM MEET PEOPLE WHO ARE WILLING TO HELP

SOLUTION 2

POV2

Climbers can "request beta" so nearby climbers get a gentle notification, prompting experienced climbers to approach and provide help, possibly building connections

PROTOTYPE

Critical Assumption:
Experienced climbers will respond to digital requests for help during their sessions.

Why critical:
We have gathered from our interviews that climbers rarely ever reject a request for help in person. However, given the annonimity of the app, people might not feel obligated to go out of their way to help. If climbers don't feel comfortable to do so, climbers requesting for help might be left helpless and isolated.

Prototype format

We conducted this prototype in person at the climbing gym. This prototype tests if experienced users would be willing to help without being asked upfront.

Materials

- Paper-based notification

Target users— Beginner/Intermediate level solo climbers who want advice from other climbers.

Procedure

Round 1 - Attempt on their own

Have a climber climb a wall they have difficulty climbing.

Round 2 - Ask for help

When they reach a point where they are stuck, ask if they would like to receive help from a climber in the gym.

Round 3 - Ask if somebody is willing to provide help

Ask a climber nearby if they would be willing to help if a request like this popped up on their phone.
If yes, bring them to the climber who needs help, and provide help.

Round 4 - Interaction

Allow them to interact and observe if it is a helpful experience.

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EXPERIENCE PROTOTYPE 2: PARTICIPANTS

Participant 1: Climber needing help

Who are they?
A female UCSB undergraduate student who is an intermediate climber.

How were they recruited?
At UCSB Recreation Centre's Climbing Gym

Why are they relevant?
They just came down from a climb after not being able to climb up and seemed like they would need help. They represent the user who were seeking for climbing community's help

Participant 2: Climber providing help

Who are they?
A male UCSB undergraduate student who is an intermediate climber.

How were they recruited?
At UCSB Recreation Centre's Climbing Gym

Why are they relevant?
He completed the same problem the climber needed help with.

EXPERIENCE PROTOTYPE 2: RESULT

What worked

- Climbers were very happy to help whenever they can
- When the climbers were working on the same problem, they were able to converse comfortable and receive help

What didn’t work

- We weren’t able to simulate the condition where climbers make the decision to help on their own accord. Since we had to explain the experiment, they might feel morally obliged to help. However, the climbers did happily mention they would help if a notification as such popped up on their phones.

Was the assumption valid?

- Yes, the climbers were all very willing to help. Climbers had a very positive reaction when presented with the request for help

New Learnings

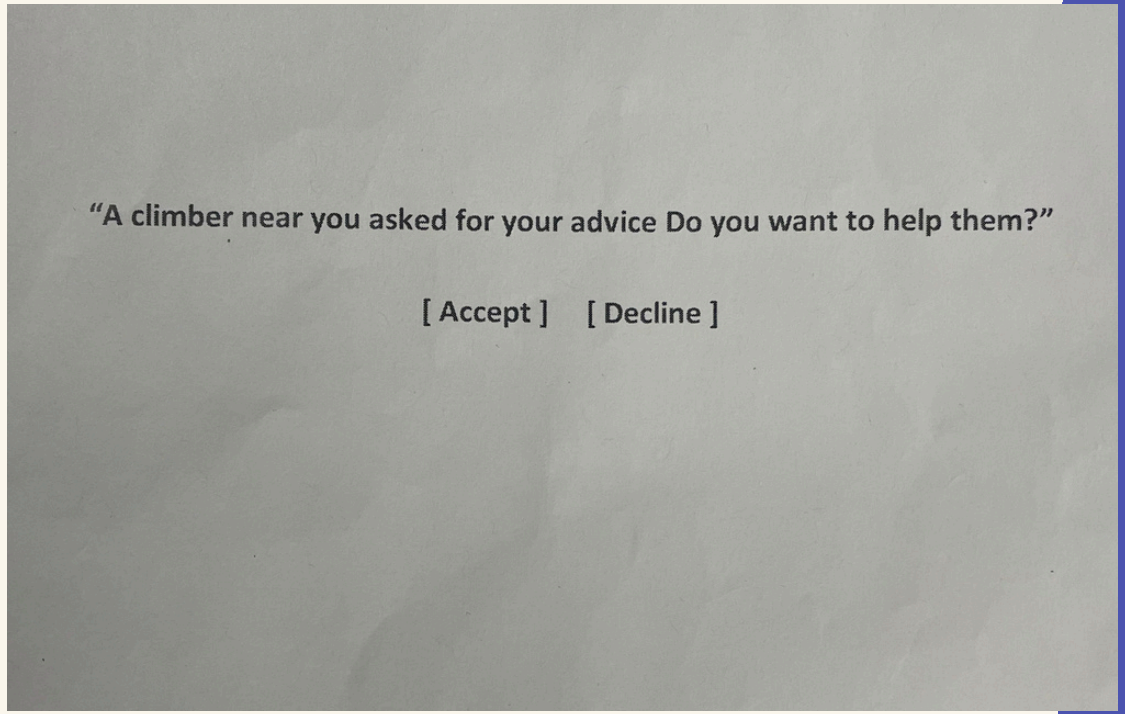
- It’s important that climbers who approach are able to help them with their climbs, otherwise it might result in an awkward position for both users

Participant 1

Participant 2

- Participant 2 helping participant 1 with the route she's struggling with

Notification



HMW3: How might we enable advanced climbers who prefer climbing alone to find creative solution for questions without disrupting their autonomy?

SOLUTION 3:

AN AI CHALLENGE MODE
THAT RESTRICTS CLIMBERS TO USING SPECIFIC HOLDS TO ATTEMPT A ROUTE,
ENCOURAGING THE DISCOVERY OF NEW SOLUTIONS ON THEIR OWN

CLIMBERS WHO STILL WANT TO HAVE OWNERSHIP OF THEIR SOLUTIONS AND WOULD PREFER TO CLIMB ALONE COULD FURTHER PROGRESS BY LEARNING NEW AND CREATIVE WAYS THEY COULD CLIMB

SOLUTION-3

POV3

An AI challenge mode that restricts climbers to using specific holds to attempt a route, encouraging the discovery of new solutions.

PROTOTYPE

Critical Assumption:
When climbers are given constraints, they will generate alternative movement strategies instead of repeating the same habits of climbing

Why critical: If constraints don't lead to new strategies (only frustration/repetition), the challenge mode provides little value.

GR2

Prototype format

Human pretending to be a AI system instructing the participant in real time.

Materials

- A Laser pointer
- Climbing route
- Path recording, Constraint rule list

Participant

Who: UCSB graduate students who are intermediate to advance level climbers.
How recruited: On site, we asked for participation
Why relevant: He is advanced level climber
Target users— Someone who want to challenge themselves with creative climbing routes.

Procedures

- Round 1 — Ask the participant to try original path (no restrictions)

- Round 2 — Restrict using some of the holds from the original route.
  - The researcher acts as the AI system.
  - The researcher points the holds that they cannot use from the original route using a laser pointer as the participant climb.

- Round 3 — Ask the participant to try finding a new route according to the restrictions.
  - The participants challenge themselves by trying to find a creative routes that they never tried before in given condition.

EXPERIENCE PROTOTYPE 3 : PARTICIPANTS

Participant 1:

Who are they?
A male UCSB graduate student who is an advanced climber

How were they recruited?
At UCSB Recreation Centre's Climbing Gym

Why are they relevant?
They are advanced and are able to think of their moves more creatively, which is also a skill they would need given that less people are able to provide help since most people are not that advanced yet.

EXPERIENCE PROTOTYPE 3: RESULT

What worked

The climber was able to climb an easier route with added difficulty by restricting the holds he could use, actively forcing him to think of other ways to solve the problem. The climber presented positive feedback for the challenge and indicated that they would use a feature like this if it existed.

What didn't work

The restricted holds were decided by us and could have been too easy. A more knowledgeable AI could generate a more challenging path and help them think more creatively.

Was the assumption valid?

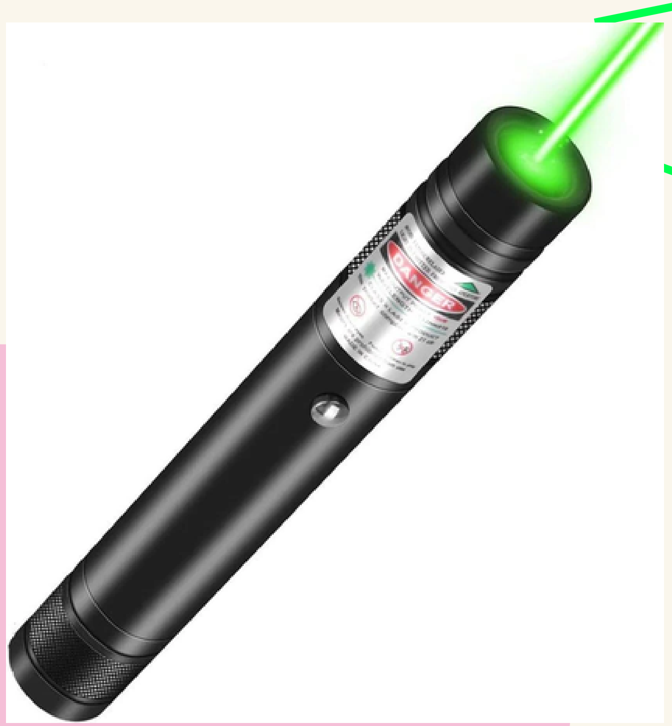
Yes, the climber indicated that they would use a feature as such to think help them think outside the box in future climbs.

New Learnings

Climbers enjoy the challenge and are always looking for ways to improve.

Participant 1

The holds we restricted



The AI Actor pointed the holds not to use

SOLUTION1

POV1

An AI-assisted system that analyzes a climber's failed attempt and suggests multiple alternative betas based on their body type, reach, and skill level.

SOLUTION2

POV2

Climbers can "request beta" so nearby climbers get a gentle notification, prompting experienced climbers to approach and provide help, eliciting conversation

SOLUTION3

POV3

A challenge mode that restricts climbers to using specific holds to attempt a route, encouraging the discovery of new solutions.

GR2

THE SOLUTION WE CHOSE

CLIMBSPACE

NEXT STEPS

WHICH SOLUTION WILL MOVE FORWARD?

A Hybrid Solution Combinging Solutions 1 & 3

- An AI-assisted beta generator that provides multiple physique-aware movement options, rather than a single "correct" solution
- A constraint-based challenge mode that introduces intentional limitations, encouraging climbers to experiment and discover alternative strategies on their own

CHOSE THIS SOLUTION BECAUSE OUR INTERVIEWS AND PROTOTYPES SHOWED THAT

- There is no one-fits-all beta
- Climbers still want autonomy while receiving guidance
- Climbers want solutions after giving it their best tries
- Exploring multiple possible paths is more helpful than being given a single answer
- Climbers are always finding ways to improve and challenge themselves

NEXT STEPS

WHO DOES IT SERVE?

- Climbers whose height or reach does not align with common beta, making standard solutions ineffective or frustrating
- Intermediate climbers who need a broader range of movement strategies to progress
- Solo climbers who prefer exploring and understanding movement solutions independently rather than relying on direct instruction

WHO MIGHT IT LEAVE OUT?

- Climbers who strongly prefer in-person coaching and direct verbal feedback
- Highly advanced climbers who can set their own betas
- Highly advanced climbers who already generate and evaluate multiple betas intuitively

ETHICAL IMPLICATIONS

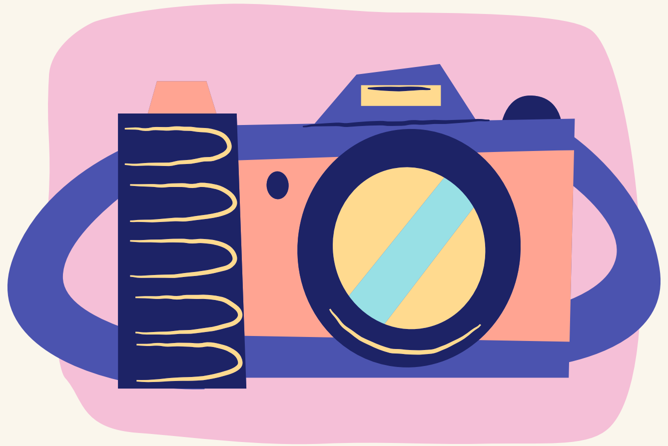
- Consent for uploading users' climbing videos and information to be analyzed by AI
- Consent of collecting and using users' data
- Be transparent about the use of LLM suggestions
- May encourage user overly-reliant on AI guidance
- May have to include safety disclaimers to prevent risky movement

GR1

THANKYOU

CLIMBSPACE

APPENDIX



GR2

USER	PROBLEM	SOLUTION
JUNE	Beta does not fit their body type	Customized solution for body type and level
REILEY	Loss of autonomy due to intrusive feedback	Constraint-based exploration
DAN	Social support needed	AI-mediated social learning

GR2

INITIAL POVS

CLIMBSPACE

GR2

POV 1 INITIAL

- **WE MET…**

  June, an 18-year-old UCSB freshman who climbs regularly and enjoys bouldering as both a social activity and a way to solve movement problems.

- **WE WERE SURPRISED TO NOTICE…**

  She often watches other people's beta and tries to follow it, but still gets stuck when those strategies don't work for her body or style.

- **WE WONDER IF THIS MEANS…**

  She needs clearer beta guidance that works better for different types of climbers.

- **IT WOULD BE GAME CHANGING TO…**

  Provide body-type-based beta suggestions for each route.

GR2

POV 2 INITIAL

- **WE MET...**

  Dan, a researcher from Belgium, who is a beginner-to-intermediate climber with one and a half years of experience. He enjoys climbing as a fun hobby instead of as an intense workout

- **WE WERE SURPRISED TO NOTICE...**

  that he enjoys the social aspect of climbing, receiving on-the-spot advice while climbing and having small talks while resting, more than just climbing itself

- **WE WONDER IF THIS MEANS...**

  having someone there to support him and talk with replenishes both his mental and physical energy to keep him going

- **IT WOULD BE GAME CHANGING TO...**

  help him sustain both mental and physical energy throughout sessions

GR2

POV 3 INITIAL

- **WE MET…**

Riley, an experienced climber and climbing instructor at the UCSB Recreation Center. She has been climbing for several years and is deeply familiar with both climbing techniques and route setting.

- **WE WERE SURPRISED TO NOTICE…**

Even though Riley enjoys climbing with others, she intentionally goes to a more distant gym in order to have time to climb alone. She approaches climbing not only as a participant, but also from the perspective of a route setter.
She does not see failure as a personal limitation, but as something influenced by body type, technique, or route design.

GR2

POV 3 INITIAL

- **WE WONDER IF THIS MEANS...**

Riley values uninterrupted focus and intentionally seeks distance from social interaction in order to think and reflect.
Not all climbers benefit from the same type of feedback.
Route setters are already aware that not all problems work for every body type or skill level.

- **IT WOULD BE GAME CHANGING TO...**

Riley could receive advice only when she chooses, without disrupting her focus.

She could understand how different types of climbers approach her routes.
She could move fluidly between the roles of instructor and learner within the same climbing space.

FULL BRAINSTORMING

All brainstorming were conducted on Google Docs together

FULL BRAINSTORMING (GOOGLE DOC USED)

BRAIN STORMING

—---

How might we help climbers explore alternative approaches when the common beta doesn’t fit their body or style?

A climbing route database filtered by body type, allowing climbers to view route strategies provided by people with different heights, arm spans, and climbing styles.

A movement mixing system where climbers select posture constraints to generate different strategies.

A multi-path route map that visualizes multiple possible climbing sequences instead of a single strategy route.

A posture difference feedback tool that highlights the differences between a climber's body posture and the strategy examples.

A strategy failure case library that collects and shares failed attempts, helping climbers understand what methods don’t work and why.

A climbing assistant that identifies a climber’s movement tendencies and recommends different strategies accordingly.

A partner matching mechanism that pairs climbers with distinctly different styles to exchange different strategy ideas.

A blind test strategy challenge that gradually reveals route hints instead of showing the complete solution.

A route planner that virtually removes or replaces holds to help climbers plan different movements.

After repeated failed attempts, the system helps categorize likely failure causes such as body position, timing, reach, or force generation, guiding users toward what to change next.

A shared library of unsuccessful attempts explains why certain approaches fail, helping climbers avoid repeating ineffective strategies.

Users can view multiple strategy attempts at the same time in a split-screen layout, making it easier to compare body positioning, sequencing, and pacing differences between climbers.

FULL BRAINSTORMING (GOOGLE DOC USED)

How might we ensure climbers feel mentally supported during sessions, even if they don't have friends to go with?

Periodic Session Check-Ins
- Checks in with users every half hour, providing encouraging messages and a chatbot for small talks

Climbing Buddy Finder
- A social app that matches climbers of similar levels to meet each other and plan to have climbs together

Rest-Time Guidance
- Suggestions during rest breaks like breathing cues, mindset resets, or reflective prompts

Progress Reflection Throughout Session
- A short recap highlighting effort, persistence, and learning to remind climbers of their overall achievements, not just focusing on successful climbs.

Real-time online support
- Climbers can seek for tips and advice for specific walls on a forum, especially from the gym staff on the floor. Gym community also shares encouraging messages

Coach Finder
- Find experienced people to provide them with encouragement and advice based on ratings from other users

Gentle Stop Signals
- Kind prompts suggesting a mental break or problem switch when frustration spikes

Virtual Climbing Buddy
- AI climbing buddy that says encouraging cheers as you climb through earphones

Post-Fail Reframes
- After a fall, offer one positive reframing like "You committed further this time"

Anonymous Encouragement Wall
- Climbers leave short supportive notes that others can see during rests

Failure Normalization Feed
- Stories or quotes from other climbers about repeated failure before success

Gym-specific communication app
- Shows how many others are also climbing alone, with a "let's climb together" indicator to join new groups or individuals

Request Beta
- Climbers can "request beta" and nearby climbers get a gentle notification, prompting experienced climbers to approach and provide help and elicit conversation

FULL BRAINSTORMING (GOOGLE DOC USED)

How might we enable climbers who prefer climbing alone to receive guidance without disrupting their focus or autonomy?

Real-Time Audio Guidance System
- Users place a camera on a tripod while climbing, and the AI provides real-time audio guidance through earphones, suggesting possible next holds or movements.

Route Image–Based Beta Generator
- Users upload an image of a climbing route, and the AI analyzes the layout to generate possible beta or movement strategies.

Skill-Level Based Community Matching
- The system analyzes a climber's skill level and connects them with others of similar experience for discussion and learning.

Body-Type–Based Partner Matching
- Based on height, reach, body type, and experience level, the system matches climbers who may benefit from sharing similar movement strategies.

Personalized Route Recommendation System
- Users input physical attributes and experience level, and the system recommends routes that are suitable or challenging for them.

Peer Feedback Exchange Platform
- Users upload climbing videos and receive feedback from other climbers through text or visual annotations.

Post-Climb Reflection Tool
- After a climbing session, users can reflect on where they struggled and why, helping them recognize patterns in their performance.

Multi-Solution Visualization
- The system shows how different climbers approached the same route, highlighting multiple valid solutions instead of a single correct one.

Movement Constraine Mode
- AI constraint challenge mode that restricts climbers to using specific movements to attempt a route, encouraging the discovery of new solutions.

Opt-In Advice Mode
- Users can choose when they want to receive guidance, allowing them to maintain autonomy and focus during climbing.

Silent Hint System
- Hints are provided visually rather than verbally, allowing climbers to receive guidance without disrupting concentration.

Route Setter Feedback Tool
- Route setters can review feedback about how climbers experienced a route and identify areas that may need adjustment.

Self-Comparison System
- Instead of competing with others, users compare current attempts with their own past performances to track progress.

Learning Pattern Tracker
- The system tracks how users approach problems over time, focusing on movement strategies rather than success or failure.

GR2

SELECTED SOLUTION 1

How might we help climbers explore alternative approaches when the common beta doesn't fit their body or style?

- Body Fit Beta
  - A climbing route database filtered by body type, allowing climbers to view route strategies provided by people with different heights, arm spans, and climbing styles.
- Constraint Mix Generator
  - A movement mixing system where climbers select posture constraints to generate different strategies.
- Multi Path Mapper
  - A multi-path route map that visualizes multiple possible climbing sequences instead of a single strategy route.
- Posture Gap Feedback
  - A posture difference feedback tool that highlights the differences between a climber's body posture and the strategy examples.
- Failure Case library
  - A strategy failure case library that collects and shares failed attempts, helping climbers understand what methods don't work and why.
- Style Style Aware Recommender
  - A climbing assistant that identifies a climber's movement tendencies and recommends different strategies accordingly.
- Style Match Partner
  - A partner matching mechanism that pairs climbers with distinctly different styles to exchange different strategy ideas.
- Progressive Hint Challenge
  - A blind test strategy challenge that gradually reveals route hints instead of showing the complete solution.
- Virtual Hold Planner
  - A route planner that virtually removes or replaces holds to help climbers plan different movements.
- Failure Cause Analyzer
  - After repeated failed attempts, the system helps categorize likely failure causes such as body position, timing, reach, or force generation, guiding users toward what to change next.
- Failed Beta Library
  - A shared library of unsuccessful attempts explains why certain approaches fail, helping climbers avoid repeating ineffective strategies.
- Beta Compare Panel
  - Users can view multiple strategy attempts at the same time in a split-screen layout, making it easier to compare body positioning, sequencing, and pacing differences between climbers.

GR2

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GR2

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- Self-Comparison System
  - Instead of competing with others, users compare current attempts with their own past performances to track progress.
- Learning Pattern Tracker
  - The system tracks how users approach problems over time, focusing on movement strategies rather than success or failure.

GR2

PROTOTYPE MATERIAL FOR SOLUTION 1 (PRE AND POST SURVEY)

Pre Solution1

1. Climbing Experience

1. How long have you been climbing?

( ) Less than 6 months

(X) 6 months – 1 year

( ) 1–2 years

( ) 3–5 years

( ) More than 5 years

2. How often do you climb?

( ) Less than once a week

(X) 1–2 times per week

( ) 3–4 times per week

( ) 5+ times per week

3. What type of climbing do you mostly do? (Select all that apply)

☒ Bouldering

☐ Top rope

☐ Lead climbing

☐ Outdoor climbing

☐ Gym only

2. Physical Profile

4. Height

5'3 cm / ft

5. Arm span (if known)

_____ cm / ft

(X) Not sure

Pre Solution1

6. How would you rate your flexibility?

(1 = Very inflexible, 5 = Very flexible)

1☐ 2☐ 3☒ 4☐ 5☐

7. How would you rate your upper body strength?

(1 = Very weak, 5 = Very strong)

1☒ 2☐ 3☐ 4☐ 5☐

8. How would you rate your footwork technique?

(1 = Very weak, 5 = Very strong)

1☐ 2☐ 3☒ 4☐ 5☐

3. Climbing Skills & Style

9. Which climbing skills are you comfortable with? (Check all that apply)

☒ Flagging

☒ Heel hook

☐ Toe hook

☐ Dyno

☒ Drop knee

☒ High step

☐ Mantling

☐ Smearing

☐ Crimping

☐ Pinching

10. What is the hardest grade you have successfully completed?

(Bouldering or route)

V5, 5.11D

Solution 1

Post-Experiment Survey

AI-Assisted Climbing Beta Exploration

1. Overall Experience

1. How helpful was the AI-generated alternative beta for understanding your failed attempt?
(1 = Not helpful at all, 5 = Extremely helpful)
☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5

2. After seeing the alternative beta suggestions, how clear was it why your original attempt did not work?
☐ Not clear at all
☐ Slightly clear
☒ Moderately clear
☐ Very clear
☐ Extremely clear

3. Compared to before, how confident do you feel about trying a new approach to the route?
☐ Much less confident
☐ Slightly less confident
☒ No change
☐ Slightly more confident
☐ Much more confident

2. Perception of the AI Feedback

4. The suggested beta felt appropriate for my body type and ability level.
☐ Strongly disagree
☒ Disagree
☐ Neutral
☐ Agree
☐ Strongly agree

5. The AI feedback helped me understand that failure does not necessarily mean lack of skill.
☐ Strongly disagree
☐ Disagree
☒ Neutral
☐ Agree
☐ Strongly agree

Solution 1

6. Which of the following best describes how you perceived the AI's role?
☐ A coach giving the "right answer"
☒ A guide helping me explore options
☐ A tool for reflection
☐ A replacement for human advice
☐ Not helpful

3. Reflection & Behavior Change

7. After this experience, how likely are you to try a different movement strategy instead of repeating the same beta?
☐ Very unlikely
☐ Unlikely
☐ Neutral
☒ Likely
☐ Very likely

8. Did this experience change how you think about failure in climbing?
☐ Yes, significantly
☐ Yes, slightly
☒ No change
☐ Not sure

If yes, how?

4. Open-ended Reflection

9. What part of the AI-generated feedback was most helpful for you?

10. What felt confusing, unrealistic, or unhelpful?

- BOX DIFFS

11. If this system were available at a climbing gym, would you use it? Why or why not?

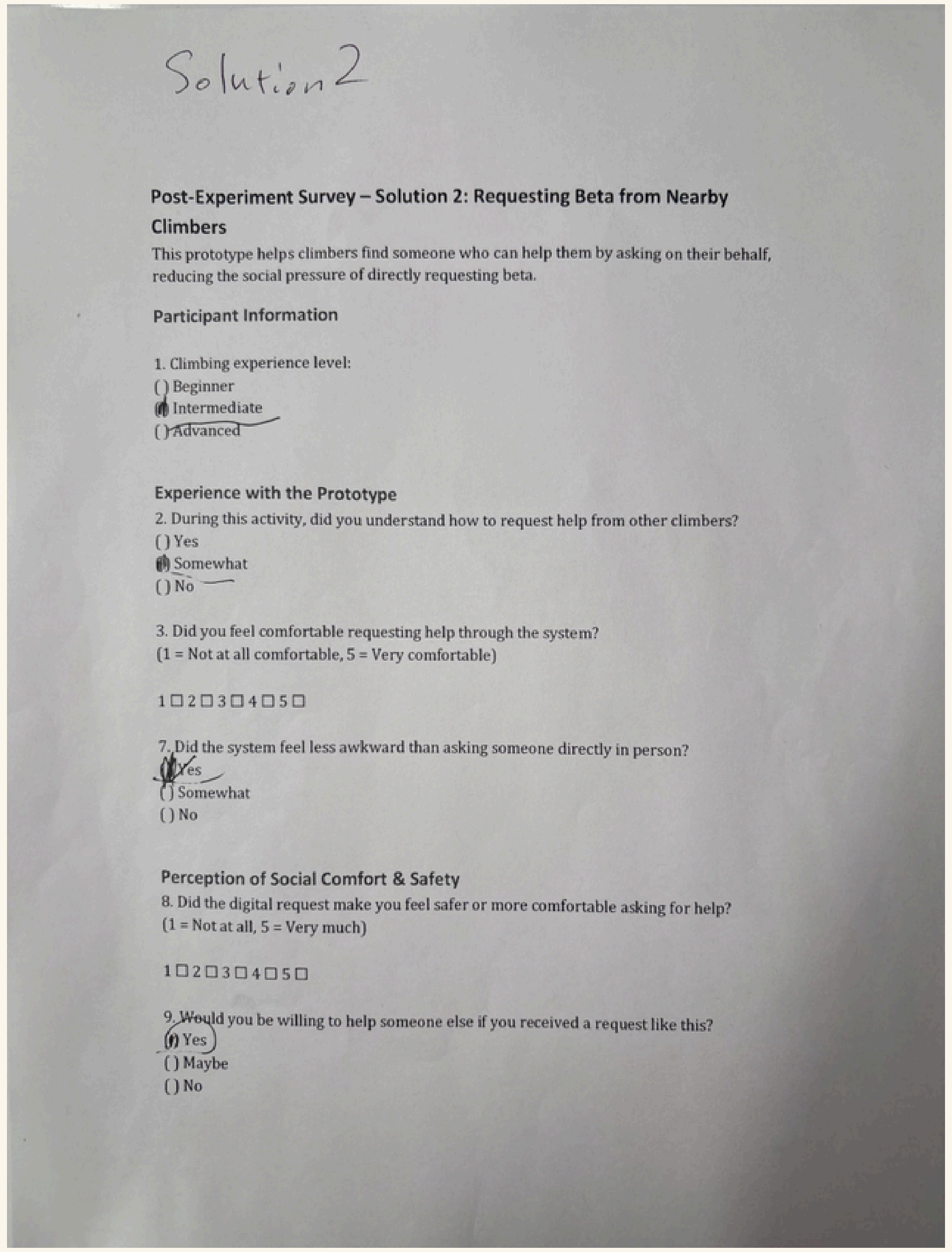
- YES, TO SEE OPTIONALITY

5. Final Evaluation

12. Overall, how effective was this prototype in helping you rethink your climbing approach?
☐ Very ineffective ☐ Ineffective ☒ Neutral ☐ Effective ☐ Very effective

GR2

PROTOTYPE MATERIAL FOR SOLUTION 2 (POST SURVEY)



Solution 2

Post-Experiment Survey – Solution 2: Requesting Beta from Nearby Climbers

This prototype helps climbers find someone who can help them by asking on their behalf, reducing the social pressure of directly requesting beta.

Participant Information

1. Climbing experience level:
( ) Beginner
(●) Intermediate
( ) Advanced

Experience with the Prototype

2. During this activity, did you understand how to request help from other climbers?
( ) Yes
(●) Somewhat
( ) No

3. Did you feel comfortable requesting help through the system?
(1 = Not at all comfortable, 5 = Very comfortable)

1☐2☐3☐4☐5☐

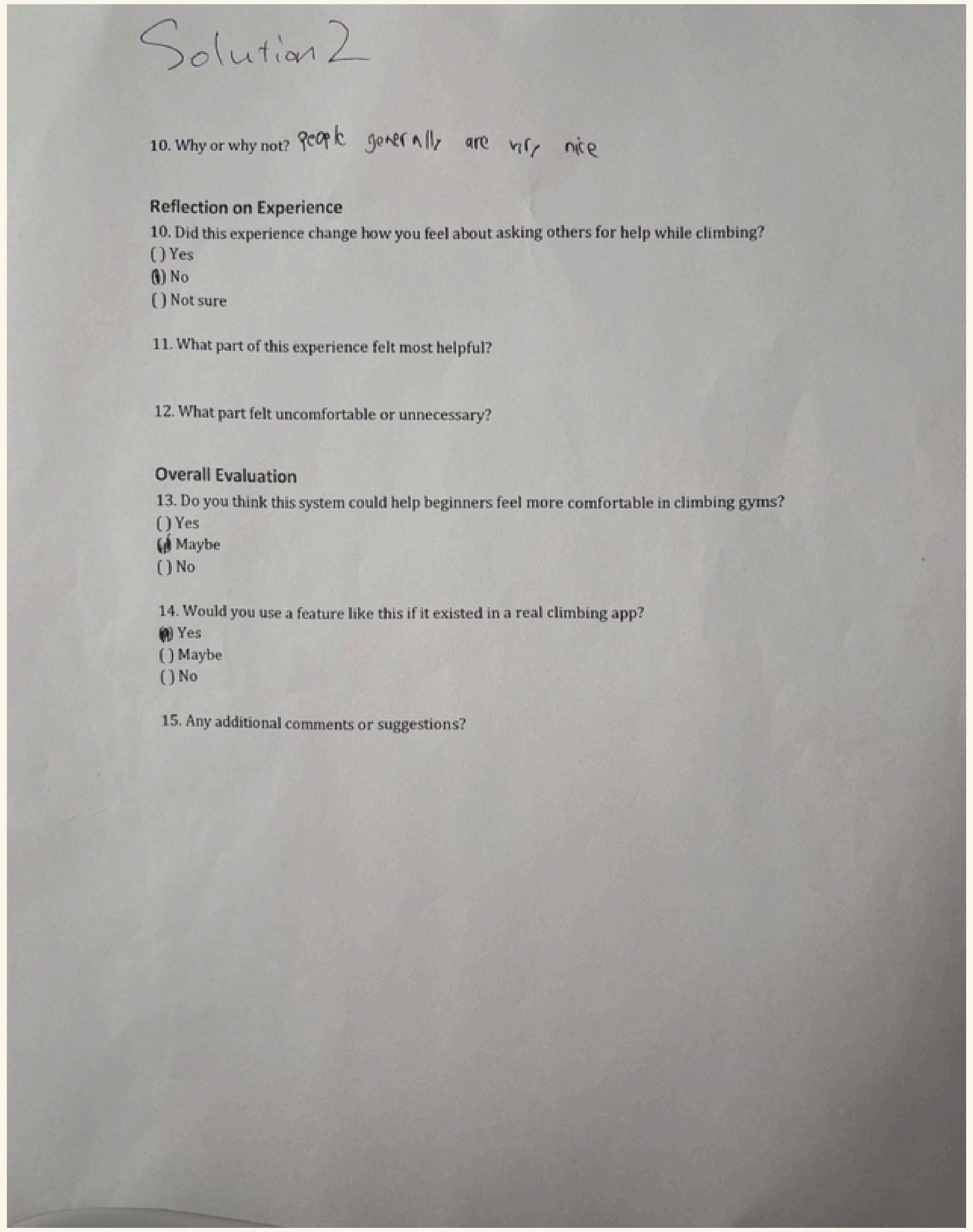
7. Did the system feel less awkward than asking someone directly in person?
(●) Yes
( ) Somewhat
( ) No

Perception of Social Comfort & Safety

8. Did the digital request make you feel safer or more comfortable asking for help?
(1 = Not at all, 5 = Very much)

1☐2☐3☐4☐5☐

9. Would you be willing to help someone else if you received a request like this?
(●) Yes
( ) Maybe
( ) No



Solution 2

10. Why or why not? people generally are very nice

Reflection on Experience

10. Did this experience change how you feel about asking others for help while climbing?
( ) Yes
(●) No
( ) Not sure

11. What part of this experience felt most helpful?

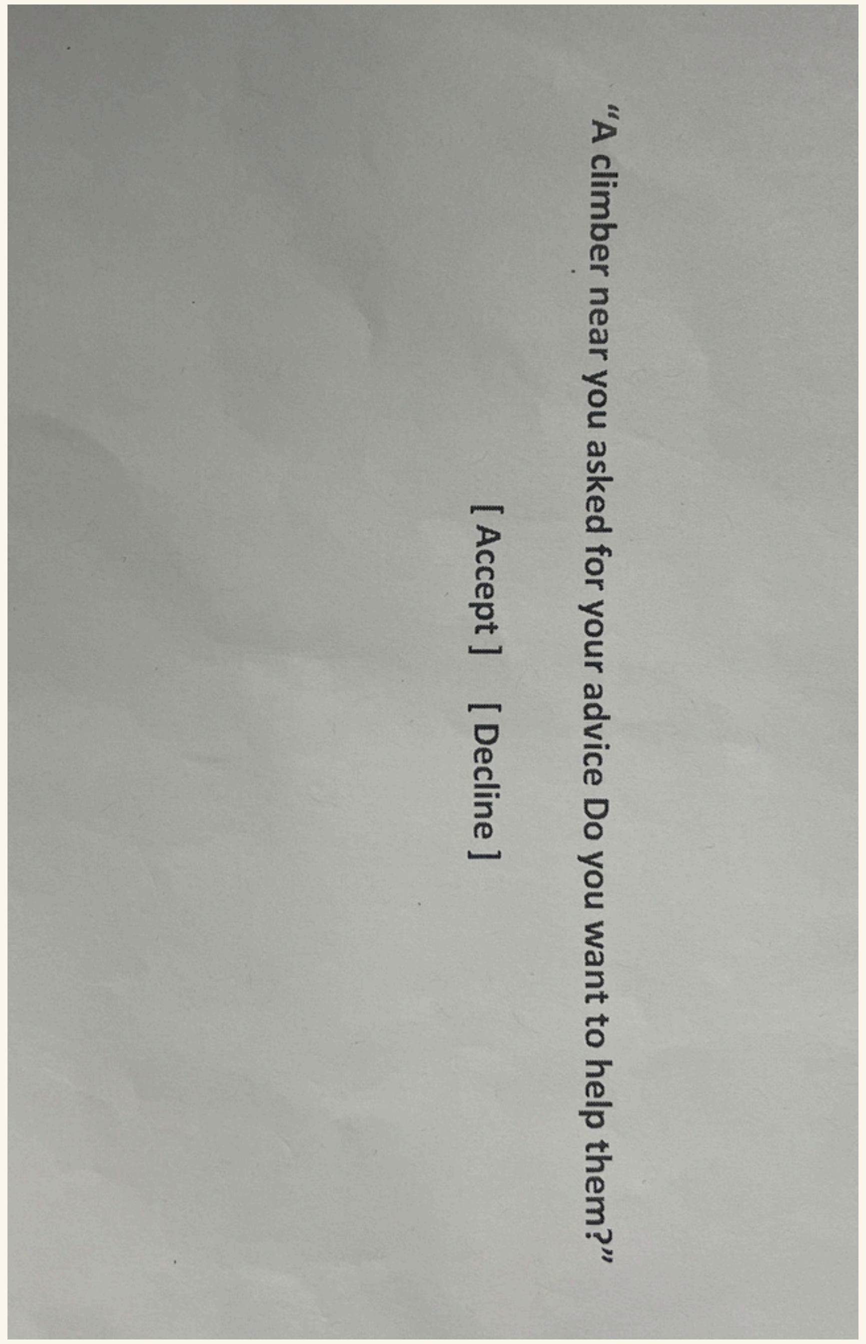
12. What part felt uncomfortable or unnecessary?

Overall Evaluation

13. Do you think this system could help beginners feel more comfortable in climbing gyms?
( ) Yes
(●) Maybe
( ) No

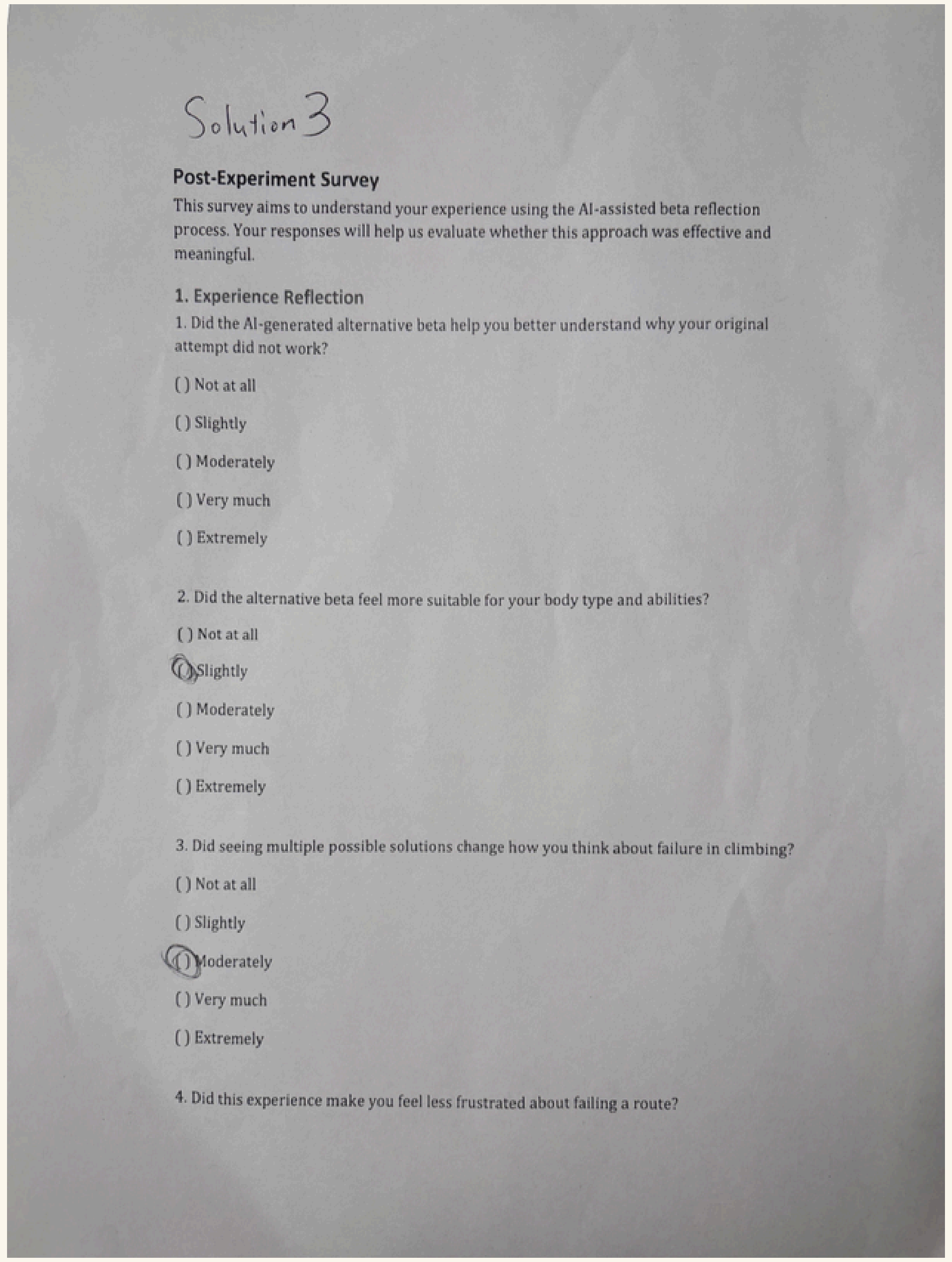
14. Would you use a feature like this if it existed in a real climbing app?
(●) Yes
( ) Maybe
( ) No

15. Any additional comments or suggestions?



GR2

PROTOTYPE MATERIAL FOR SOLUTION 3 (POST SURVEY)



Solution 3

Post-Experiment Survey

This survey aims to understand your experience using the AI-assisted beta reflection process. Your responses will help us evaluate whether this approach was effective and meaningful.

1. Experience Reflection

1. Did the AI-generated alternative beta help you better understand why your original attempt did not work?

( ) Not at all

( ) Slightly

( ) Moderately

( ) Very much

( ) Extremely

2. Did the alternative beta feel more suitable for your body type and abilities?

( ) Not at all

( ) Slightly

( ) Moderately

( ) Very much

( ) Extremely

3. Did seeing multiple possible solutions change how you think about failure in climbing?

( ) Not at all

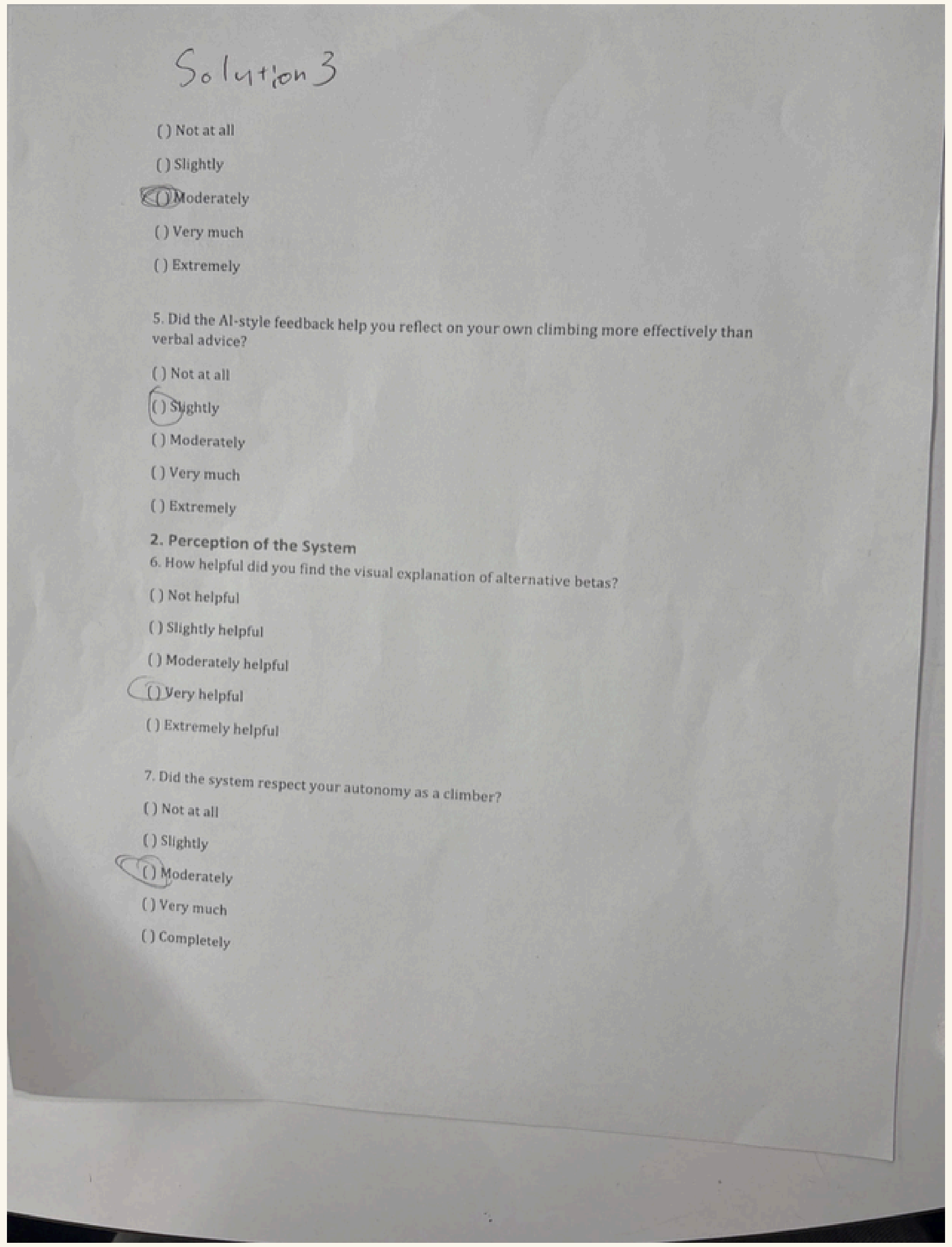
( ) Slightly

( ) Moderately

( ) Very much

( ) Extremely

4. Did this experience make you feel less frustrated about failing a route?



Solution 3

( ) Not at all

( ) Slightly

( ) Moderately

( ) Very much

( ) Extremely

5. Did the AI-style feedback help you reflect on your own climbing more effectively than verbal advice?

( ) Not at all

( ) Slightly

( ) Moderately

( ) Very much

( ) Extremely

2. Perception of the System

6. How helpful did you find the visual explanation of alternative betas?

( ) Not helpful

( ) Slightly helpful

( ) Moderately helpful

( ) Very helpful

( ) Extremely helpful

7. Did the system respect your autonomy as a climber?

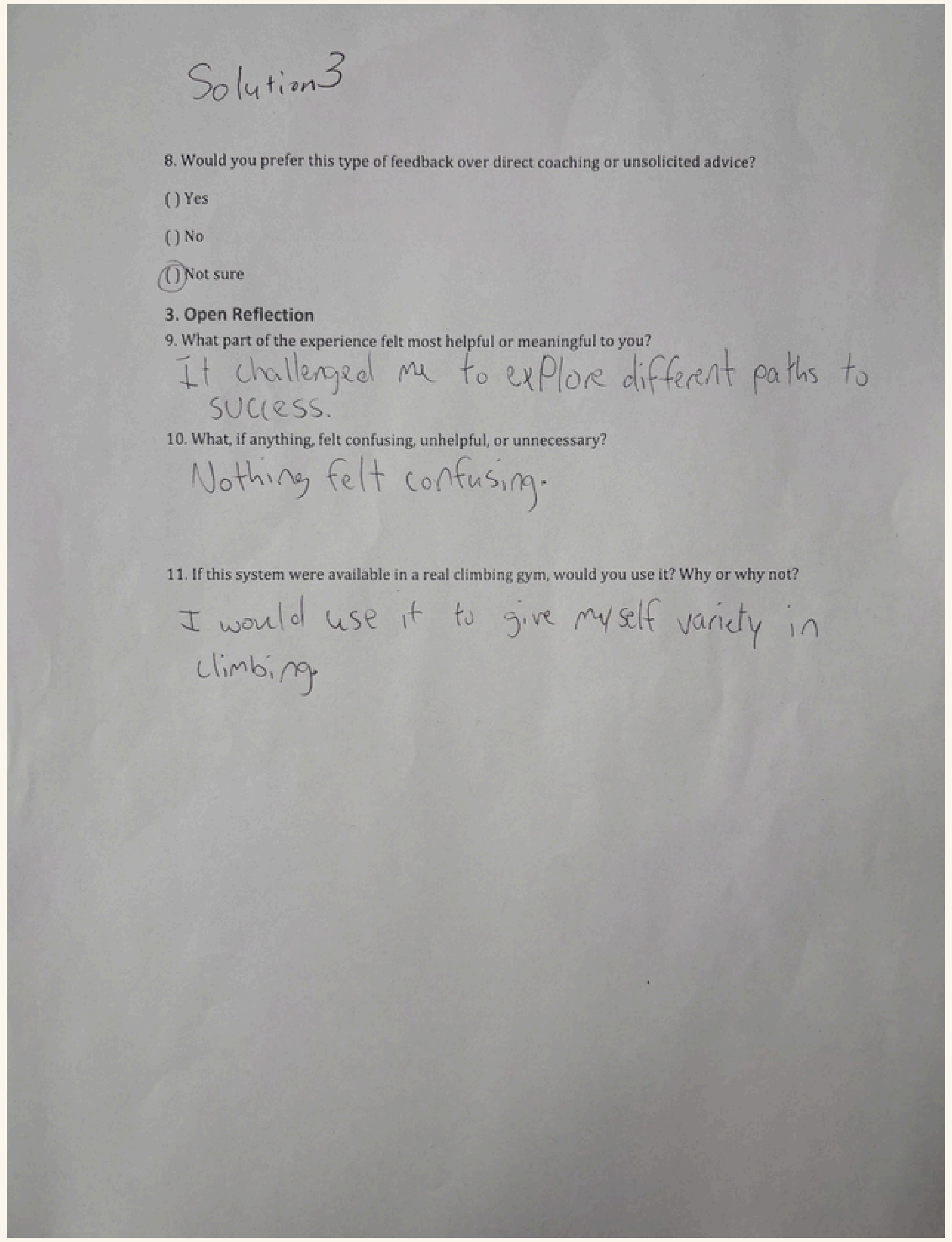
( ) Not at all

( ) Slightly

( ) Moderately

( ) Very much

( ) Completely



Solution 3

8. Would you prefer this type of feedback over direct coaching or unsolicited advice?

( ) Yes

( ) No

( ) Not sure

3. Open Reflection

9. What part of the experience felt most helpful or meaningful to you?

It challenged me to explore different paths to success.

10. What, if anything, felt confusing, unhelpful, or unnecessary?

Nothing felt confusing.

11. If this system were available in a real climbing gym, would you use it? Why or why not?

I would use it to give myself variety in climbing.

IMPROVEMENTS ACCORDING TO PEER FEEDBACK

- We made sure our inferences and key insights were deeper and less observational
- We interviewed climbers who prefered to climb alone, which was a popular suggestion from our peers
- We included a small terminology explanation to allow readers to understand the common terminologies used in climbing
- We included new questions to be asked in the interview, such as:
    - How do climbers emotionally interpret getting stuck (do they see it as failure, challenge, or normal progression) and how does that interpretation shape their motivation to continue?
    - when someone says they want progress visibility, what exact kind of progress do they mean, and what would make it feel motivating instead of stressful?
- We narrowed down our real target user group to decide on the optimal solution